BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, AJK, MIRPUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: 22/01

Retyped faithfully from the original board paper — same questions, no changes.

MATHEMATICS (PART –II) (OBJECTIVE PART)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 22/01

Note: Note:- Write your Roll No. in space provided. Over writing, cutting, using of lead pencil will result in loss of marks. All questions are to be attempted.

Q1. (Marks (20)) Each question has four possible answers, Tick (√) the correct answer.

Q1. $x = a\cos\theta$, $y = a\sin\theta$ are parametric equations of a;

(A) Circle
(B) Parabola
(C) Ellipse
(D) Hyperbola

Q2. $\lim_{x\to a} (x^n - a^n)/(x - a) =$

(A) na^n
(B) na^{n+1}
(C) na^{n-1}
(D) $(n+1)a^{n-1}$

Q3. If $f(x) = \tan^{-1}x$, then $f'(x) =$

(A) $-1/(1-x^2)$
(B) $1/(1-x^2)$
(C) $-1/(1+x^2)$
(D) $1/(1+x^2)$

Q4. If $y = \ln(\sin x)$, then $dy/dx =$

(A) $1/\sin x$
(B) $\tan x$
(C) $\cot x$
(D) $\sec x$

Q5. If $f'(c) = 0$, then $f(x)$ has relative maxima at $x = c$ if;

(A) $f''(c) = 0$
(B) $f''(c) < 0$
(C) $f''(c) > 0$
(D) $f''(c) \neq 0$

Q6. (1/2) d/dx ($\tan^{-1}x - \cot^{-1}x$) =

(A) $1/(1+x^2)$ (B) $1/(1-x^2)$

(C) $-1/(1+x^2)$ (D) $-1/(1-x^2)$

Q7. $\int a^x\,dx$ =

(A) $a^x + c$ (B) $a^x.\ln a + c$

(C) $(1/\ln a).a^x + c$ (D) $(\ln a)/a^x + c$

Q8. $\int e\hat{}(\tan x).\sec^2 x\,dx$ =

(A) $\tan x + c$ (B) $e\hat{}(\sec^2 x) + c$

(C) $e\hat{}(\tan x).\tan x + c$ (D) $e\hat{}(\tan x) + c$

Q9. $\int e\hat{}(-x)[-\sin x + \cos x]\,dx$ =

(A) $e\hat{}(-x)\sin x + c$ (B) $e\hat{}(-x)\cos x + c$

(C) $e\hat{}(-x)\tan x + c$ (D) $e\hat{}(-x)\cot x + c$

Q10. The solution of the differential equation $dy/dx = \sec^2 x$, is;

(A) $y = \cot x + c$ (B) $y = \ln x + c$

(C) $y = \cos x + c$ (D) $y = \tan x + c$

> Note: (P.T.O)

Q11. Slope of the line parallel to $3x - 4y + 5 = 0$, is;

(A) 4/3 (B) 3/4

(C) 3/5 (D) 5/3

Q12. The distance of the point P(4,−6) form x-axis is;

(A) 4 (B) −4

(C) 6 (D) −6

Q13. The lines represented by $ax^2 + 2hxy + by^2 = 0$, are real and distinct if;

(A) $h^2 - ab < 0$ (B) $h^2 - ab = 0$

(C) $h^2 - ab \neq 0$ (D) $h^2 - ab > 0$

Q14. Slope of horizontal line is;

(A) Zero (B) ∞

(C) 1 (D) −1

Q15. $x = 4$ is the solution of the inequality;

(A) $-2x + 3 > 0$ (B) $x + 3 > 0$

(C) $x - 3 < 0$ (D) $x + 3 < 0$

Q16. The radius of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, is;

(A) $g^2 + f^2 + c$ (B) $g^2 + f^2 - c$

(C) $\sqrt{(g^2 + f^2 + c)}$ (D) $\sqrt{(g^2 + f^2 - c)}$

Q17. Length of latus-rectum of the parabola $(y - 2)^2 = -4(x - 7)$, is;

(A) 4 (B) -4

(C) 8 (D) -8

Q18. Vertices of the ellipse $x^2/2 + y^2/9 = 1$, are;

(A) $(\pm 3, 0)$ (B) $(0, \pm 3)$

(C) $(0, \pm\sqrt{2})$ (D) $(\pm\sqrt{2}, 0)$

Q19. Projection of $a = i - k$ along $b = j + k$ is;

(A) $-1/\sqrt{2}$ (B) $1/\sqrt{2}$

(C) $1/2$ (D) $-1/2$

Q20. The vectors u, v and w are coplanar if;

(A) $u.v \times w = 0$ (B) $u.v \times w = 1$

(C) $u.v \times w = 2$ (D) $u.v \times w = 3$

> Note: (The End)

MATHEMATICS PAPER : PART – II (SUBJECTIVE PART)

Time Allowed: 2:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 22/01

> Note: Note:- Attempt any Twenty Five (25) short questions in all selecting eight from Q. 2 and Q. 3 each and nine from Q. 4.

> Note: SECTION – I

Q2. (Marks (2 x 8 = 16)) Write short answers of any eight parts.

(i) Find $(f(a+h) - f(a))/h$ and simplify where $f(x) = 6x - 9$

(ii) If $f(x) = 1/\sqrt{(x-1)}$, $g(x) = \sqrt{(x^2+1)}$ Find (a) fog(x) (b) gof(x)

(iii) Find Lim $(x \to +\infty)$ $(2-3x)/\sqrt{(3+4x^2)}$

(iv) Find Lim $(x \to 0)$ sin ax / sin bx

(v) Find Lim $(x \to \infty)$ $(x/(1+x))^x$

(vi) Find the derivative of $1/\sqrt{x}$ w.r.t 'x' by definition.

(vii) If $y = (2x-1)/\sqrt{(x^2+1)}$ find dy/dx

(viii) Find dy/dx if $y^2 + x^2 - 4x = 5$

(ix) If $y = x \cos y$, find dy/dx

(x) Find $f'(x)$, if $f(x) = \text{Sin}^{-1}\sqrt{(1-x^2)}$

(xi) Find dy/dx, if $y = (\ln \tanh x)$

(xii) If $y = \sqrt{x} + 1/\sqrt{x}$, find y_2

Q3. (Marks (2 x 8 = 16)) Write short answers of any eight parts.

(i) Solve the differential equation $\sin y \operatorname{cosec} x \, dy/dx = 1$

(ii) Find area bounded by the curve $y = \sqrt{(3-x)}$ and above x-axis form $x = -1$ to $x = 2$

(iii) Evaluate $\int_1^2 (x^2+1)\, dx$

(iv) Evaluate $\int x^3 \ln x \, dx$

(v) Evaluate $\int \cos(\sqrt{x} - x/2) \times (1/\sqrt{x} - 1)\, dx$

(vi) Evaluate $\int \sqrt{y}(y+1)/y \, dy$

(vii) Find δy and dy if $y = \sqrt{x}$ when x changes from 4 to 4.41

(viii) Evaluate $\int dx/(\sqrt{(x+1)} - \sqrt{x})$

(ix) Find an equation of a line passing through $(-4,7)$ and parallel to the line $2x - 7y + 4 = 0$

(x) Express the lines $3x + 4y - 7 = 0$, $2x - 5y + 8 = 0$, $x + y - 3 = 0$ in matrix form and check whether the lines are concurrent.

(xi) Find the two lines represented by $6x^2 - 19xy + 15y^2 = 0$

(xii) Find area of the region bounded by the triangle whose vertices are $(a, b + c)$, $(a, b - c)$ and $(-a, c)$

Q4. (Marks (2 x 9 = 18)) Write short answers of any nine parts.

(i) Define objective function for linear programming problems.

(ii) Graph the solution set of $3x + 7y \geq 21$ linear inequality in xy-plane.

(iii) Find the centre and radius of the circle with given equation $x^2 + y^2 - 6x + 4y + 13 = 0$

(iv) Determine whether the point $(-5,6)$ lies outside, on or inside the circle $x^2 + y^2 + 4x - 6y - 12 = 0$

(v) Find the length of the tangent drawn from the point $(-5,4)$ to the circle $5x^2 + 5y^2 - 10x + 15y - 131 = 0$

(vi) Write an equation of the parabola with focus $(2,5)$ and directrix $y = 1$

(vii) Find foci of the ellipse $x^2 + 4y^2 = 16$

(viii) Find slope of the tangent to the curve $x^2/a^2 + y^2/b^2 = 1$ at $(a \cos \theta, b \sin \theta)$

(ix) If $u = 2i - 7j$, $v = i - 6j$ and $w = -i + 5$ find $(1/2)u + (1/2)v + (1/2)w$

(x) Find a unit vector in the direction of $v = i + 2j - k$

(xi) Find a real number α so that the vectors $u = \alpha i + 2\alpha j - k$ and $v = i + \alpha j + 3k$ are perpendicular.

(xii) Compute the cross product $a \times b$ and $b \times a$ when $a = 3i - 2j + k$, $b = i + j$

(xiii) Find cosine of the angle θ between $u = i - 3j + 4k$, $v = 4i - j + 3k$

> Note: (P.T.O)

> Note: (2)

> Note: SECTION – II

> Note: Note:- Attempt any three questions from this section. All questions carry equal Marks. (10x3=30)

Q5. (Marks (10)) (a) If $f(x) = \{ (\sqrt{(2x+5)} - \sqrt{(x+7)})/(x-2), x \neq 2; k, x = 2 \}$. Find value of k so that f is continuous at x = 2 (05)

Q. (Marks (05)) (b) Differentiate ab-initio w.r.t. 'x' sin √x

Q6. (Marks (10)) (a) Evaluate $\int 2x/(x^6 - 1)\, dx$ (05)

Q. (Marks (05)) (b) The points A(−1,2), B(6,3) and C(2,−4) are vertices of a triangle. Show that the line joining the midpoint D of AB and the midpoint E of AC is parallel to BC and DE = (1/2)BC

Q7. (Marks (10)) (a) Solve the differential equation; $y - x\, dy/dx = 3(1 + x\, dy/dx)$ (05)

Q. (Marks (05)) (b) Graph the solution region of the system of linear inequalities and find the corner point. $3x + 7y \leq 21$, $2x - y \leq 3$, $y \geq 0$

Q8. (Marks (10)) (a) Find the interior angles of the triangle whose vertices are A(2,−5), B(−4,−3), C(−1,5) (05)

Q. (Marks (05)) (b) Show that the circles $x^2 + y^2 + 2x - 2y - 7 = 0$ and $x^2 + y^2 - 6x + 4y + 9 = 0$ touch externally.

Q9. (Marks (10)) (a) Find an equation of the ellipse with foci (±√5,0) and passing through the point (3/2, √3) (05)

Q. (Marks (05)) (b) Prove that the points whose position vectors are A(−6i + 3j + 2k), B(3i − 2j + 4k), C(5i + 7j + 3k), D(−13i + 17j − k) are coplanar.

> Note: (The End)